

- The **Mid Term Exam 2- Part I** for **STA 2023-0002** will be held in class from **10:30 AM to 11:45 AM** on **April 16, 2026**.
- The exam will cover Chapters 4-6.
- The exam is **closed-book and closed-notes**.
- You may use a **calculator** and **two double-page cheat sheet**.
- You are **not allowed** to use laptops, cellphones, tablets, or any other electronic devices.
- All cellphones must be **turned off** before the exam begins.
- Show all necessary steps for full credit.
- **Write all work clearly.** Answers without supporting steps may not receive credit.
- **Write answers in the spaces provided.** If you need additional space, clearly label the continuation.
- **Total points: 60.**

By signing below, I acknowledge that I have read and understood the instructions above and agree to comply with the university's policies on Academic Integrity for this exam.

Academic Integrity Statement: "I acknowledge that I have fully complied or will comply with all aspects of the university's policies on Academic Integrity and the exam instructions in submitting this exam."

Name: _____

Student ID: _____

Answer all questions. Show all necessary steps for full credit.

1. Sample Space and Classical Probability

A fair coin is flipped 3 times.

- (a) List the sample space.
- (b) Find the probability of getting exactly 2 heads.
- (c) Find the probability of getting at least 1 tail.

2. Complement Rule and Multiplication Rule

A hospital receives 8 blood pressure monitors, and 2 are defective. Two monitors are selected at random without replacement.

- (a) What is the probability that both selected monitors are not defective?
- (b) What is the probability that at least one defective monitor is selected?

3. Contingency Table, Conditional Probability, and Independence

The table below shows the marital status of 200 adults in a small community.

Marital Status	Male	Female	Total
Never married	30	20	50
Married	40	50	90
Widowed	5	15	20
Divorced	10	20	30
Separated	4	6	10
Total	89	111	200

Answer the following:

- (a) What is the probability that a randomly selected person is female?
- (b) What is the probability that a randomly selected person is widowed?
- (c) What is the probability that a randomly selected person is widowed and female?
- (d) Given that the person selected is female, what is the probability that the person is widowed?
- (e) Let

A = the person is widowed, B = the person is female.

Are events A and B independent? Show work and justify your answer.

4. Counting Techniques: Permutations and Combinations

A club has 12 members.

- (a) In how many ways can a president, vice president, and secretary be selected?
- (b) In how many ways can a committee of 3 members be selected?

5. Discrete vs. Continuous Random Variables

For each of the following variables, state whether it is **discrete** or **continuous**.

- (a) The number of students absent from class on a given day.
- (b) The weight of a randomly selected watermelon.
- (c) The number of flat tires a driver gets in one year.
- (d) The amount of time it takes a student to finish an exam.

6. Mean of a Discrete Random Variable

The random variable X represents the profit (in dollars) from a game and has the following probability distribution:

x	-2	0	3	5
$P(x)$	0.25	0.40	0.20	0.15

- (a) Find the mean or expected value of X .
- (b) Interpret the mean in context.

7. Binomial Probability

A student guesses on 8 true/false questions. Let X be the number of questions answered correctly.

- (a) Why is this a binomial experiment? State the values of n , p , and q .
- (b) Find the probability that the student gets exactly 6 correct.
- (c) Find the probability that the student gets fewer than 3 correct.
- (d) Find the probability that the student gets at least 3 correct.

8. Mean and Standard Deviation of a Binomial Random Variable

A basketball player makes free throws with probability 0.80. Suppose the player attempts 20 free throws. Let X be the number of successful free throws.

- (a) Find the mean of X .
- (b) Find the standard deviation of X .
- (c) Interpret the mean in context.

9. Finding Areas Under a Normal Curve

SAT verbal scores are approximately normally distributed with mean

$$\mu = 500$$

and standard deviation

$$\sigma = 100.$$

Let X represent the score of a randomly selected student.

- (a) Find the probability that a student scores less than 620.
- (b) Find the probability that a student scores more than 450.
- (c) Find the probability that a student scores between 480 and 560.

10. Finding the Value of a Normal Random Variable

The weights of newborn babies are approximately normally distributed with mean

$$\mu = 7.4$$

pounds and standard deviation

$$\sigma = 1.2$$

pounds.

- (a) Find the weight that corresponds to the 30th percentile.
- (b) Find the two weights that separate the middle 80% of baby weights from the lowest 10% and highest 10%.